

# TOWER OF HANOI PUZZLE SOLVER IN JAVASCRIPT



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You can solve the Tower of Hanoi puzzle using this JavaScript program. Tower of Hanoi is a mind game that's counted among puzzles like Sudoku and Rubik's cube. This game has its origin from the Indian sub-continent and then spread all over the world. Even though it was born in India, it became popular in Vietnam, for its shape is like a Pagoda, which is most commonly seen in China, Vietnam and such south-east Asian countries; hence the natives named it on the lines of their country capital Hanoi.

## The puzzle

Tower of Hanoi puzzle has various disks or rings of different circumferences. These disks can be placed in such a way that the smaller disk is placed on top of the larger disk, arranged just like a stack. Some or all of these disks are coloured for easy recognition. The disks are placed in an increasing order from top, i.e., the smallest disk is placed on the top of the stack and the largest one at the bottom. The program has in total three

such towers, of which two are empty. The one having all the disks on the left-hand side is the source (S). The other



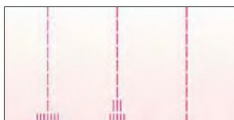
Screenshot 1: Initialising three disks on tower 'S'



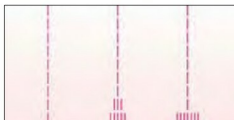
Screenshot 2: Moving topmost smaller disk to tower 'D'



Screenshot 3: Moving the other larger disk to tower 'I'



Screenshot 4: Moving the smaller disk from tower 'D' to tower 'I'



Screenshot 5: Moving the larger disk from tower 'S' to tower 'D'

in the middle is the intermediate (I) and the tower onto the right side is the destination (D).

The rules of the puzzle are:

1. One should move only one disk at a time.
2. A larger disk should not sit on a smaller disk.

The puzzle is said to be successfully solved if all the disks on tower 'S' are arranged onto tower 'D' in the same order as it was on tower 'S' with the help of tower 'I' by following the above rules.

Each of the disks follows a binary formula for successful completion of the tower (see the table).

## Program description

The program, written in JavaScript, is an automatic 'Tower of Hanoi' puzzle solver that helps one to solve the puzzle by just following the steps as shown in the program output screenshots. It can generate disks automatically from 1 to 9 and solves the puzzle step-by-step. JavaScript is used as a multifaceted program code that solves problems with much ease.

The element that the page used in this code is a button that has been rendered using the <style> tag. The <style> tag can be used to apply various styles to the element. The button element has been modified via the <style> element by a class name 'Hanoi'.

In the main program code, a few variables are declared for maximum height of the tower (9). The timer parameter is required for automatic operation of the puzzle. An array of 511 moves (refer the table) is declared for source 'S' as well as destination 'D'. Poles are then declared as an array of three, one each for source, intermediate and destination poles. A spacer is also declared to keep each of these towers

## Various Combinational Disks and Moves Required to Get the Solution

| No. of disks (n) on 'S' | No. of moves (y) | Formula   |
|-------------------------|------------------|-----------|
| 1                       | 1                |           |
| 2                       | 3                |           |
| 3                       | 7                |           |
| 4                       | 15               |           |
| 5                       | 31               | $y=2^n-1$ |
| 6                       | 63               |           |
| 7                       | 127              |           |
| 8                       | 255              |           |
| 9                       | 511              |           |